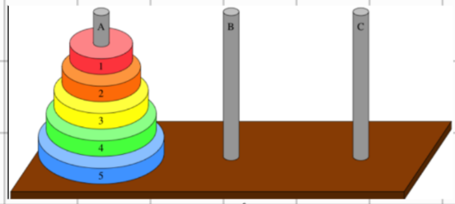
CS 173 - Lecture 22

NP

- P - Polynomial, can be solved in polynomial time (eg. $o(1)$, $o(n)$, $O(n\log n)$, $O(n^{10000})$, etc.)
- EXP - Exponential time, 2^n, 3^n, etc. can be solved in exponential time (eg. tower of hanoi, parse trees of a sentence)

ex. Tower of hanoi



Start Goal Temp

Trying to move tower to another stick

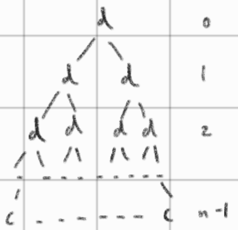
To move tower of n levels, need to first move n-1 tower

Recursive complexity:

$T(1) = c$

$T(n) = 2T(n-1) + d$

Recursion tree:

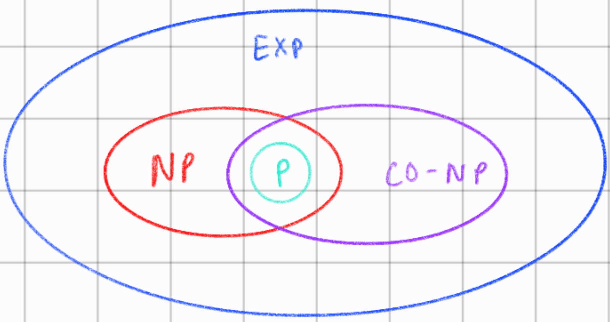


$o(2^n)$ (see lecture 21 for better breakdown of recursion tree)

- Some problems haven't been proved to be able to go faster, so ∃ some problems that can be verified in polynomial time. But best solution might take longer, which is NP
- Verified yes - provided w/ answer, can check if correct ≠ finding answer (diff. time complexities)

eg. sorting algo, verified yes would be determine if sorted, solving would be sorting an actual unsorted list

- Unknown if all problems in NP can be solved in polynomial time
- Currently, best soln. takes exp. time but no proof that polynomial solution doesn't exist
- Co-NP - Can be verified no in polynomial time



- ∃ problems ∉ P but ∈ NP and ∈ CO-NP

eg. given integer n, factor as product of primes

NP Complete

- Best solution now is exponential, but no proof can't be solved in polynomial
- Problems are intertransformable
- If we find polynomial solution to anything in NP-Complete, we can show that P=NP